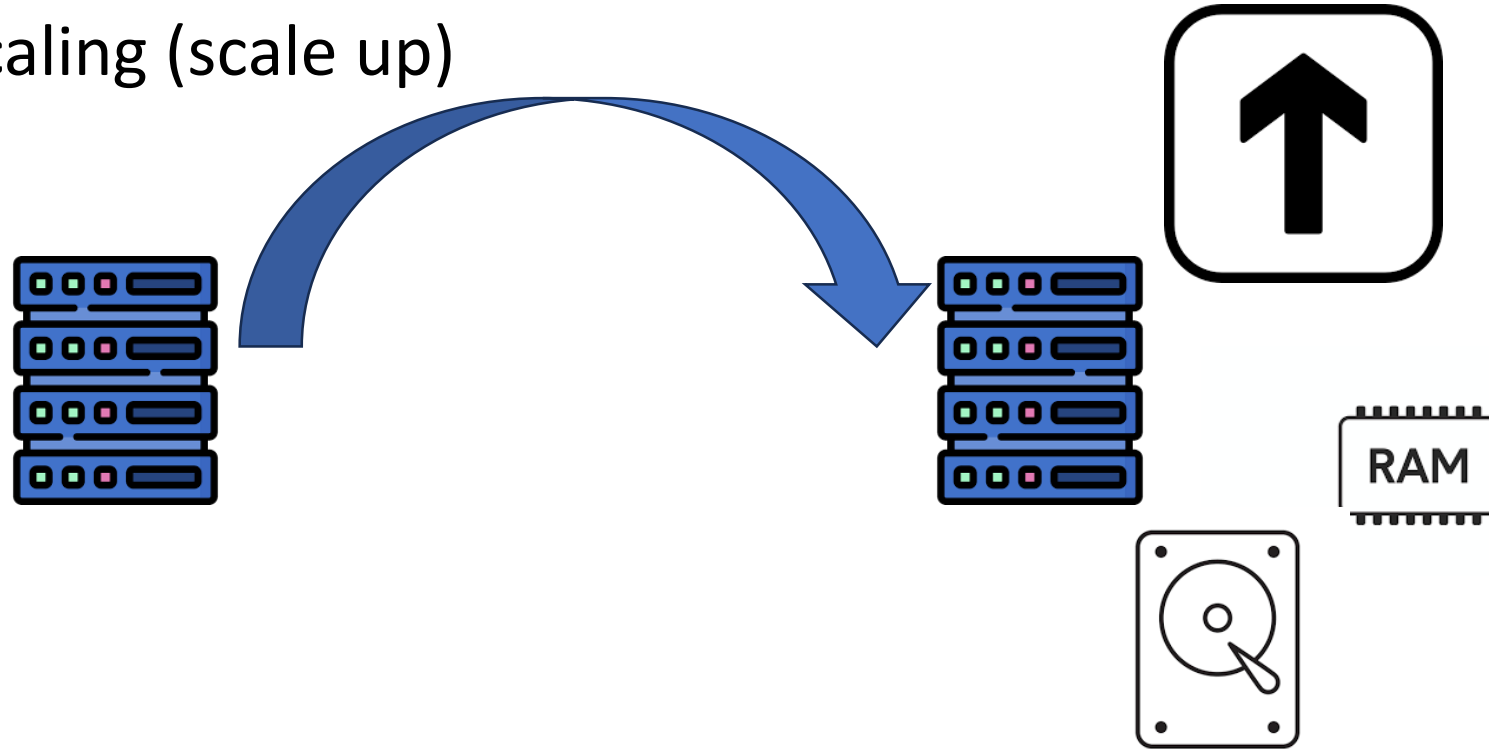


Vertical Vs Horizontal Scaling

Prepared By: Dr. D. P. Singh

Vertical and Horizontal scaling

- Vertical scaling (scale up)



Increases a single server's capacity by **upgrading hardware components** like CPU, RAM, or storage.

Pros and Cons of Vertical Scaling

Pros:

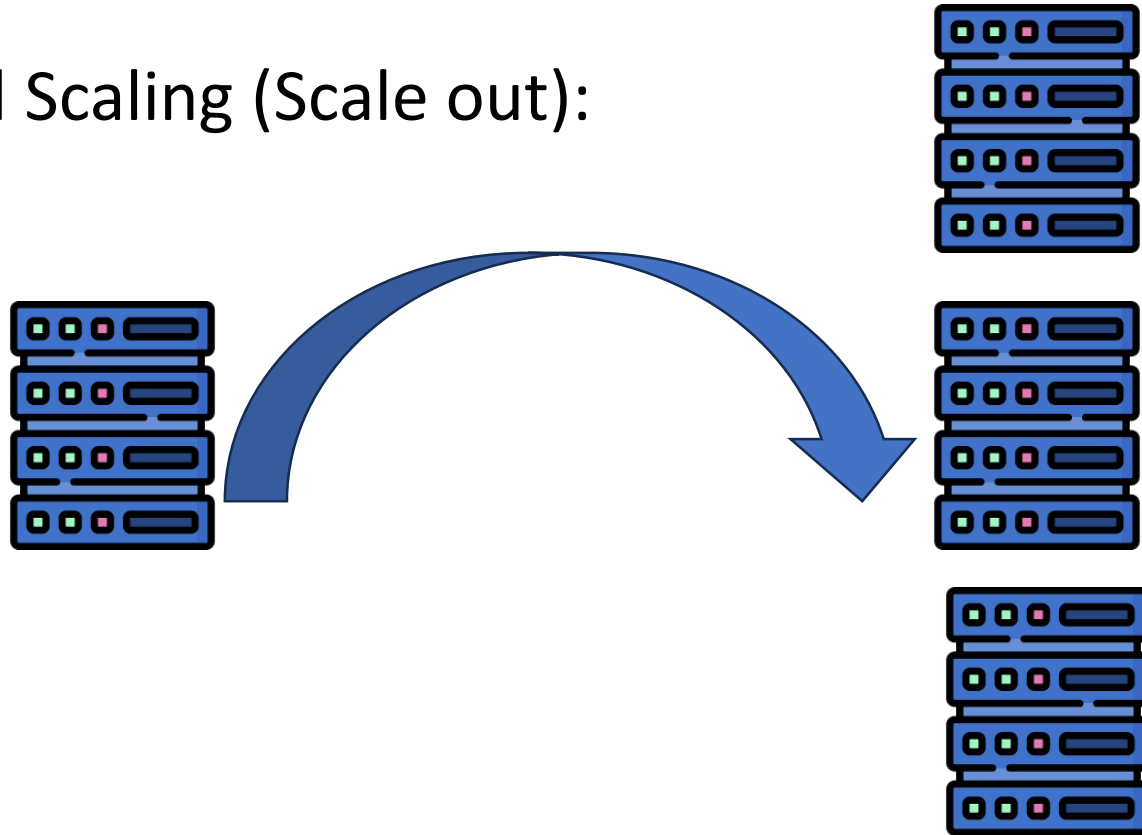
- **Simplicity:** No application code changes or complex clustering required.
- **Performance:** High performance for a single node.
- **Data Consistency:** Easier to manage because data resides in one place.

Cons:

- **Hard Ceiling:** Limited by the maximum, physical capacity of the hardware.
- **Downtime:** Often requires shutting down the server for upgrades.
- **Single Point of Failure (SPOF):** If the server fails, the entire application goes down.
- **Cost:** High-end, customized hardware can be very expensive.

Horizontal Scaling

- Horizontal Scaling (Scale out):



Increases system capacity by **adding more machines or nodes** to a network rather than upgrading existing hardware

Pros and Cons of Horizontal Scaling

Pros

- **Unlimited Scalability:** Theoretically limitless, allowing systems to grow indefinitely by adding more commodity hardware.
- **High Availability & Resilience:** Distributes workloads across multiple nodes, ensuring that if one fails, the system stays online.
- **No/Low Downtime:** Systems can be scaled up or down on the fly without interrupting service.
- **Cost-Effective Long-Term:** Uses affordable, commodity hardware rather than specialized, expensive, high-end servers

Pros and Cons of Horizontal Scaling

Cons

- **Implementation Complexity:** Requires distributed architecture, load balancers, and **complex application design**.
- **Data Consistency Challenges:** Managing data across multiple machines can **lead to consistency issues** (e.g., latency, synchronization).
- **Higher Initial Setup Costs:** Higher upfront investment for infrastructure, configuration, and networking setup.
- **Management Overhead:** Maintaining, monitoring, and upgrading numerous servers is **more operationally intensive**.

Thank You